Mykhailo Pyvovarov

Senior Software Engineer | Python | AI Agents | SaaS Platforms

Hirka Polonka, Ukraine

PROFILE

Senior Software Engineer with 10+ years of experience building scalable web platforms, AI-driven systems, and high-performance SaaS applications.

Hello! I'm Mykhailo, a senior software engineer specializing in modern web development and AI-powered automation. Over the past decade, I've helped startups, SaaS companies, and enterprises design and build robust digital products — from full-stack platforms to intelligent AI agent systems that automate workflows and drive business growth.

My expertise combines traditional software engineering with cutting-edge AI automation. I design systems that integrate LLMs, AI agents, and workflow automation into real-world applications used by thousands of users.

I focus on delivering clean architecture, scalable systems, and production-ready solutions that help businesses move faster and operate smarter.

I prioritize long-term partnerships, clean code, and delivering measurable results for my clients.

If you're looking for a reliable senior engineer who understands both software architecture and modern AI technologies, I'd love to collaborate with you.

SKILLS

Programming
Python, JavaScript, TypeScript, SQL

Frontend Development
React, Next.js, JavaScript, HTML5, CSS3, TailwindCSS

Backend Development
Django, FastAPI, Node.js, REST API Development, Microservices Architecture, GraphQL

Databases
PostgreSQL, MySQL, Redis, Vector Databases, Chroma Databases

Cloud & DevOps
Docker, CI/CD Pipelines, Cloud Infrastructure, Containerized Deployment

AI & Automation
AI Automation, AI Agents, LLM Integration, LangChain, RAG Systems

PROFESSIONAL EXPERIENCE

Innovative AI Systems Inc
Senior AI & Full Stack Engineer
04/2021 – 10/2025 | New York, United States

- Led the development of AI-powered SaaS automation platforms integrating large language models, AI agents, and workflow orchestration systems.
- Designed and implemented LLM-based intelligent assistants capable of automating customer support, knowledge retrieval, and internal business processes.
- Built scalable backend architectures using Python (FastAPI, Django) and microservices, enabling high-availability AI services used by thousands of users.
- Developed AI agent frameworks capable of interacting with APIs, databases, and external services to automate complex multi-step workflows.
- Integrated vector databases and embedding pipelines to enable semantic search and retrieval-augmented generation (RAG) applications.
- Designed cloud-native architectures deployed with Docker and scalable infrastructure supporting AI workloads.
- Implemented monitoring systems and evaluation pipelines to track AI model performance, reliability, and response quality in production.
- Collaborated with product managers and engineering teams to translate business requirements into scalable AI platform features.
- Optimized system performance through asynchronous processing, caching strategies, and efficient data pipelines.

EDUCATION

Master of Computer Science(MSCS)
Taras Shevchenko National University of Kyiv
09/2012 – 06/2014 | Kyiv, Ukraine

Bachelor of Science in Computer Science
Taras Shevchenko National University of Kyiv
09/2008 – 06/2012 | Kyiv, Ukraine

PROJECTS

Scalable SaaS Platform for Business Workflow Management
Project Goal
Designed and developed a multi-tenant SaaS platform that allows companies to manage internal workflows, tasks, and operational processes from a centralized dashboard.

Challenges & Solutions
The main challenge was designing a scalable system capable of supporting multiple organizations with isolated data environments. I implemented a multi-tenant architecture with modular backend services using Python and REST APIs.
To ensure performance under high traffic, asynchronous task queues and optimized database queries were introduced. The platform also integrated AI-based workflow suggestions, allowing the system to recommend task automation based on user behavior and historical patterns.
The result was a scalable SaaS application used by multiple organizations to automate daily operational workflows.

Real-Time Data Analytics Dashboard Platform
Project Goal
Develop a web-based analytics platform that enables companies to visualize and monitor real-time operational data through interactive dashboards.

Challenges & Solutions

Nextgen Software
Senior Software Engineer
10/2017 – 03/2021 | Bucharest, Romania
- Designed and developed large-scale SaaS platforms for enterprise and startup clients using modern full-stack technologies.
- Built scalable backend services using Python, Node.js, and REST APIs, supporting high-traffic applications and distributed systems.
- Implemented microservices architecture to improve system modularity, scalability, and maintainability.
- Developed data processing systems for analytics platforms handling large operational datasets.
- Built modern web interfaces using React and JavaScript frameworks for dashboards, analytics tools, and enterprise platforms.
- Integrated third-party APIs including payment systems, CRM platforms, and external data providers.
- Implemented authentication systems using OAuth, JWT, and secure access controls.
- Participated in full software development lifecycle including architecture design, development, testing, and deployment.
- Worked in Agile development environments contributing to code reviews, sprint planning, and technical documentation.

Digital Product Studio LLC
Full Stack Developer
06/2014 – 08/2017 | Seattle, WA, United States
- Built modern web applications and SaaS products for startups and growing companies.
- Designed scalable backend systems using Python and Node.js.
- Developed REST APIs supporting web and mobile applications.
- Built frontend applications using React and JavaScript frameworks.
- Implemented database systems using PostgreSQL and optimized queries for performance.
- Integrated cloud services and third-party APIs to extend product capabilities.
- Created internal dashboards and data visualization tools for business operations.
- Collaborated with designers and product teams to deliver high-quality user experiences.

TechWave Inc
Software Developer
11/2012 – 05/2014 | Pomona, California, United States
- Developed backend systems and web applications supporting high-traffic business platforms.
- Built RESTful APIs and backend services using modern web frameworks.
- Designed relational database schemas and optimized SQL queries for performance.
- Contributed to frontend development using JavaScript and web technologies.

Handling large datasets while maintaining real-time responsiveness was a major challenge. I designed a backend data processing pipeline that aggregated and cached data efficiently before delivering it to frontend dashboards.
The frontend interface was built using React and modern visualization libraries, providing customizable dashboards for different user roles.
AI-powered analytics modules were later integrated to automatically detect anomalies and highlight unusual business trends, helping teams identify issues faster.

AI-Enhanced Customer Support Platform
Project Goal
Develop a customer support system that combines traditional ticket management with AI-assisted responses to improve support efficiency.

Challenges & Solutions
The platform needed to handle thousands of support tickets while helping support teams respond faster. I built a scalable backend using Python microservices and REST APIs integrated with CRM systems.
An AI component was integrated to suggest responses based on previous tickets and company documentation, allowing support agents to resolve issues faster while maintaining human oversight.
The system significantly improved response times and reduced manual workload for support teams.

Enterprise Web Application for Internal Operations
Project Goal
Build a secure enterprise web application used internally by organizations to manage operational data, reporting, and internal collaboration.

Challenges & Solutions
The platform needed to support complex business workflows and strict access control requirements. I implemented role-based authentication systems and modular backend services using Django and PostgreSQL.

- Assisted with application deployment, debugging, and production issue resolution.
- Worked closely with senior engineers to improve application architecture and scalability.

To improve productivity, I integrated AI-powered features that could summarize internal reports and generate insights from operational data, allowing management teams to quickly review key information.

Smart Content Management System (AI-Assisted CMS)

Project Goal

Develop a modern content management system that allows businesses to manage websites, digital assets, and marketing content from a centralized platform.

Challenges & Solutions

Traditional CMS systems often lack automation and intelligent content tools. I built a scalable CMS platform using Node.js and React with a modular architecture supporting plugins and integrations.

AI features were introduced to assist with content generation, SEO optimization suggestions, and automated content tagging, helping marketing teams improve productivity.

AI-Powered Marketing Automation Platform

Project Goal

Create a marketing automation system that enables companies to manage campaigns, track performance, and automate marketing workflows.

Challenges & Solutions

The platform integrated multiple marketing tools including email systems, CRM platforms, and analytics providers. I designed scalable backend services capable of processing campaign data and user engagement metrics.

AI-powered modules were added to analyze campaign performance, generate marketing insights, and recommend optimization strategies. This helped marketing teams make faster data-driven decisions.